XBRAIN CAPSTONE · TASK FORCE 4 · AIO-03

Foresight Lens

A capacity-exhaustion prediction engine for tier-1 cloud services.

Lead time ≥ 15 min | Inference ~$0 cost | Model ewma-stl-v1

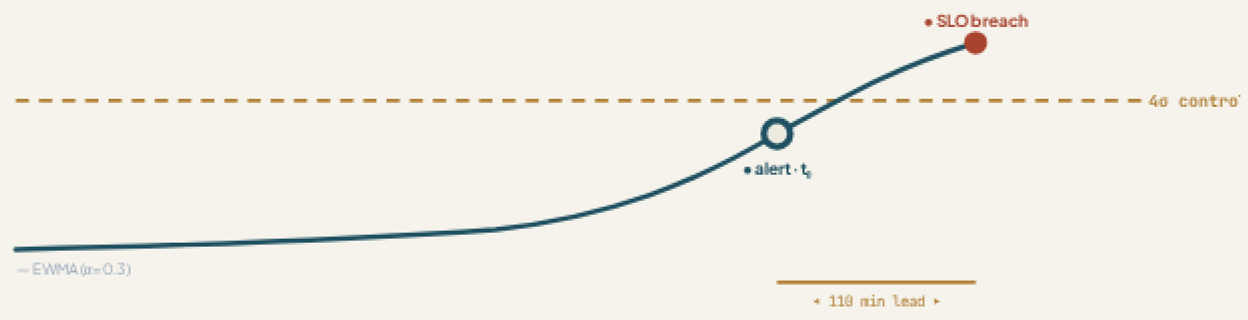

Fig. 1 — EWMA control chart raises an alert ~110 min before the de-seasonalised signal breaches the SLO.

02

CLIENT CONTEXT

When the monitor turns red, it's already too late

Reactive monitoring fires at 99% — no lead time for SRE.

CLIENT CONTEXT

Hard constraints from the client

> *"When the monitor turns red, the system is **already down**. We need to predict the incident early enough for SRE to intervene — but we don't want the AI to act on its own, and the monthly bill must stay **under $200**."*
>
> — Client interview · W11

≤ $200
TOTAL MONTHLY SPEND

Advise
NO AUTO-REMEDIATION

3 svc
PAYMENT · FRAUD · LEDGER

03

PROBLEM

Reactive monitoring alerts too late

The signal is visible for 90+ minutes, yet the threshold triggers at the cliff edge.

PROBLEM

Four forces driving the need for prediction

01 · Slow exhaustion, sudden collapse

Memory leak → OOM, CPU starvation, connection-pool & queue backlog.

02 · Reactive, not predictive

Thresholds fire at 99% — no lead time to scale up or restart.

03 · AIOps tools are heavy & costly

Market platforms blow past the $200 budget at this scale.

04 · Auto-remediation is distrusted

Fear of an AI shutting down healthy servers by mistake.

04

OUTCOMES

Predict early, advise clearly — never act alone

An advisor, not an actor. The CDO platform owns every action.

OUTCOMES

Four measurable outcomes

110	3	$0	5
MIN LEAD TIME	TENANTS ISOLATED	TOKEN COST	ACTION VERBS
Flag capacity exhaustion ≥ 15 min before SLO breach.	Per-service baselines, zero cross-tenant bleed.	In-house statistical model — no LLM tokens.	Action, target, confidence & evidence per alert.

05

HOW IT WORKS

STL + EWMA

De-seasonalise, then detect sustained drift.

ALGORITHM

Two-step detection — offline STL + real-time EWMA

STEP 1 · STL offline

Learn each service's daily load curve from clean days; store seasonal profile + residual σ. Removes the season that fools naïve detectors.

STEP 2 · EWMA inference

Run a control chart on the residual. Sustained drift breaches the K=4σ limit → anomaly. One-off spikes are smoothed away.

< 10 ms

COMPUTE / CALL

100%

DETERMINISTIC & EXPLAINABLE

DECISION

Why not an LLM?

For infrastructure time-series, exact math beats generated text.

APPROACH	COST / MO	LATENCY	FP RATE	VERDICT
STL + EWMA control chart	~$36	<10 ms	7.1%	√ chosen
Isolation Forest contamination 0.02	~$150	~20 ms	21.4%	FP gate fail
LLM (Bedrock) per-window prompt	~$10k	~2 s	—	cost + hallucination

ADR-001 · ADR-004 — a 4σ breach is a sentence any engineer can defend.

ARCHITECTURE

Stateless, one-directional request path

CDO Platform → POST /v1/predict → FastAPI + Pydantic → STL + EWMA Engine → Decision Router → RECOMMENDATION

S3 Baselines

Per-service baselines (KMS-encrypted). Read each request.

Audit Trail

SHA-256 input hash → CloudWatch / S3. 6-field log.

SRE Gate

Manual approval — humans keep the decision. No auto-remediation.

CONTRACTS

Three signed contracts — frozen W11

01 · Telemetry

- 7 gauges · CPU, memory, latency, pool…
- 1-min granularity, RFC3339 UTC
- Mandatory `tenant_id` · no PII

02 · AI API

- `POST /v1/predict`
- SLA: p99 < 500 ms · 100 RPS · 99.5%
- Errors 400 / 401 / 422 / 429 / 503

03 · Deployment

- ECS Fargate · 0.5 vCPU / 1 GB
- min 2 / max 4 · canary 10 → 50 → 100
- $200 cost circuit breaker

Schema unchanged W11 → W12 — CDO never rewrites code.

EVIDENCE

Evaluation results — measured, not hard-coded

0.971
RECALL · GATE ≥ 0.80 √

7.1%
FP RATE · GATE ≤ 12% √

110 m
LEAD TIME · GATE ≥ 15M √

0.049
BRIER · < 0.10 √ CALIBRATED

0.793
PRECISION · STRETCH ≥ 0.75 √

9/9
PYTEST SCENARIOS PASS

A/B · same holdout

STL + EWMA: .971 · FP 7.1%
Isolation Forest: .638 · FP 21.4%

Confusion matrix · 3 svc

TP 169 **FN** 5
FP 44 **TN** 574

Measured on ~790 sliding windows, 3 tier-1 services. Re-run: `tf4_evidence.py`

FINOPS

Cost model — ~$36/month

~$36

/ MONTH

Fargate 2-task HA, flat. Measured — not a forecast.

18%

BUDGET USED

Alert at $160. Circuit breaker at $200 → scale to zero.

$0

INFERENCE TOKENS

No LLM in the loop.

<$1

MARGINAL / TENANT

Scales to thousands.

Budget priority: $200 cap outranks 99.5% SLA — fail-open, CDO falls back to rules.

SAFETY

Six guardrails — defense in depth

01 · GATE

Confidence gating — Below 0.7 → downgrade SCALE_UP to INVESTIGATE.

02 · ASSIST-ONLY

Never executes — Recommends only; no closed-loop auto-remediation.

03 · AUDIT

6-field audit log — SHA-256 input hash, KMS-encrypted, no raw PII.

04 · FAIL-OPEN

Graceful degrade — 5xx → CDO falls back to rule-based alerts.

05 · ISOLATION

Per-request scoping — X-Tenant-Id enforced; stateless — zero data bleed.

06 · RATE LIMIT

Noisy-neighbour guard — 600 req/min/tenant → 429 with backoff.

ADRS

Six architecture decision records

ADR-001

Statistical model over LLM — Exactness & $0 tokens vs. natural-language RCA.

ADR-002

Confidence threshold 0.7 — Between 0.5 (too loose) and 0.8 (too strict).

ADR-003

Manual silence for flash sales — Reliable 99% of the year; no blunted sensitivity.

ADR-004

STL+EWMA over Isolation Forest — Proven by measured A/B: better recall & FP.

ADR-005

Weekly + drift-triggered retrain — Manual cadence; no auto-pipeline in scope.

ADR-006

EWMA $\alpha = 0.3$, $K = 4.0$ — From a 16-point sweep. K=3 fails FP, K=4.5 costs recall.

WHAT WE SHIPPED

All client gates met, with margin

A working FastAPI engine on stateless Fargate, with reproducible evidence — ready for W12 final build.

~$36	3	9/9	110 min
18% OF BUDGET	CONTRACTS SIGNED	TESTS PASS	LEAD TIME

APPENDIX

Backup slides

EWMA parameter sweep

Α	K	FP%	RECALL
0.3	3.0	17.5	.977
0.3	**4.0**	**7.1**	**.971**
0.3	4.5	6.5	.966
0.15	4.0	11.8	.977

Chosen α=0.3, K=4.0 — best FP margin while holding recall.

Telemetry · 7 signals

- `cpu_usage_percent`
- `memory_usage_percent`
- `api_latency_ms`
- `queue_depth`
- `active_connections`
- `db_connection_pool_pct`
- `cache_hit_rate_pct`

4 baselined · 3 z-score fallback. 1-min, 90-day retention.

API error codes

CODE	MEANING
`400`	tenant/data mismatch · no retry
`401`	missing X-Tenant-Id / auth
`422`	schema · <120 points
`429`	rate limit · backoff
`503`	unavailable → CDO fallback